

Matheus Gomes Cordeiro

Data Science | Artificial Intelligence

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LinkedIn | github.com/fobos123deimos | Personal Portfolio

Professional Summary

Data scientist with an M.Sc. in Teleinformatics Engineering from the Federal University of Ceará and experience in risk model validation, time-series forecasting, and applied artificial intelligence. Financial-sector work spans data analysis, challenger models, and proposed model governance frameworks. Combines Python, SQL, and Databricks experience with deep learning and scientific computing research, software development, and university teaching.

Technical Skills

Programming and data: Python, SQL, R, C/C++, pandas, NumPy, PySpark, and Databricks.

Machine learning and AI: scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow/Keras, Transformers/Hugging Face, and natural language processing.

Statistics and forecasting: model validation, credit risk, time-series analysis, statsmodels, sktime, NeuralForecast, and Prophet.

Software and infrastructure: Git, GitHub/GitLab, pytest, PostgreSQL, MongoDB, RabbitMQ, and Elasticsearch.

Scientific computing and visualization: Numba, Cython, mpmath, MATLAB/Octave, Mathematica, Matplotlib, Plotly, and Streamlit.

Professional Experience

University of Fortaleza (UNIFOR)

Jul 2026 – Present

Assistant Professor | Part-time

- Teach undergraduate courses in Mathematics for Computing, Problem Modeling with Mathematics, Logical and Algorithmic Reasoning, and Collaborative Cyber-Physical Systems.
- Design lectures, assessments, and projects connecting mathematics, programming, and hardware; supervise hands-on activities with ESP32, sensors, actuators, Wi-Fi, MQTT, and ESP-NOW.

ACT Digital

Aug 2025 – 2026

Data Scientist II | Client: Banco BMG

- Developed forecasts of credit-loss provisions for bank budgeting, using monthly expected credit loss (ECL) percentages.
- Led the technical design of a proposed reusable credit model validation framework, defining scoring criteria and mapping results to accepted, accepted with reservations, or rejected governance outcomes.
- Advised on an AI agent MVP for testing credit models against validation criteria, in collaboration with Domo.
- Applied statistical and machine learning methods to time-series forecasting, including ARIMA/SARIMA, Prophet, N-BEATS, N-HiTS, and XGBoost; processed and analyzed data using SQL, PySpark, and Databricks.

Technologies: Python, R, SQL, Databricks, PySpark, scikit-learn, statsmodels, and NeuralForecast.

VOYAGER IT Innovative Solutions

Dec 2024 – Jul 2025

Data Scientist II | Client: Banco do Nordeste do Brasil

- Validated **four models** within the Model Compliance Center, with work spanning credit, microcredit, operational, market, social, environmental, and climate risks.
- Performed exploratory data analysis, preprocessing, and model replication; developed challenger models and benchmarked predictive performance.
- Reviewed model documentation against internal policies and governance requirements; applied supervised learning, clustering, and statistical methods, including XGBoost, LightGBM, MLP, K-means, and Value at Risk (VaR).

Technologies: Python, R, SQL, pandas, NumPy, scikit-learn, and TensorFlow.

Entrepreneurship, Research & Development

SeaFortress

Jan 2018 – Present

Co-founder & Chief Artificial Intelligence Officer (CAIO)

- Lead the application of AI to independent games and custom virtual experiences, combining a game studio with a software development business.
- Contributed to **Unity Sentis**; a SeaFortress game was showcased at **GDC 2024**.

Technologies: Python, PyTorch, Transformers/Hugging Face, C#, Unity, and Sentis. [Company website](#).

Quantum Optics and Information Group (GIQ)

Feb 2023 – Feb 2025

Quantum Computing Simulation Researcher

- Researched numerical simulation of the quantum harmonic oscillator, focusing on accuracy and computational efficiency for photonic quantum computing; co-advised by Dr. Scott Glancy (NIST).
- Developed **Fast Wave**, a Python package with tests and technical documentation. [Repository](#) | [Documentation](#).

Technologies: Python, Numba, Cython, mpmath, pytest, pdoc, and Git.

Insight Data Science Lab

Jan 2022 – Jan 2023

Artificial Intelligence Researcher

- Applied BERT-based models to next-location prediction in vehicle trajectories, adapting natural language processing techniques to mobility data.
- Developed the **Teach** dashboard as an output of the research. [Project repository](#).

Technologies: Python, PyTorch, Transformers/Hugging Face, pandas, pymove, and Voilà.

LEAD Dell

Jul 2021 – Jan 2022

Artificial Intelligence Developer

- Contributed to an R&D team building a system-log anomaly detection application for Dell systems using Isolation Forest.
- Integrated Python-based processing into a pipeline ingesting logs from RabbitMQ and storing results in Elasticsearch.

Education

M.Sc. in Teleinformatics Engineering

Feb 2023 – Feb 2025

Federal University of Ceará (UFC), Brazil. Research in scientific computing, computational physics, and quantum system simulation.

B.Sc. in Computer Engineering

Jan 2017 – Jan 2023

Federal University of Ceará (UFC), Brazil.

Publications

Cordeiro, M. et al. (2024). Efficient computation of the wave function $\psi_n(x)$ using Hermite coefficient matrix in Python.

7th Workshop-School on Quantum Computing and Information (WECIQ), pp. 56–60.

Cordeiro, M. et al. (2022). Towards smart farming: Fog-enabled intelligent irrigation system using deep neural networks.

Future Generation Computer Systems, 129, 115–124.

Cruz, L. A. et al. (2022). Modeling Trajectories Obtained from External Sensors for Location Prediction via NLP Approaches.

Sensors, 22(19), 7475. Co-author: Matheus Cordeiro.

Freitas, N. A. de et al. (2021). Using deep learning for trajectory classification.

ICAART, vol. 2, pp. 664–671. Co-author: Matheus Cordeiro.

Cordeiro, M. G. et al. (2019). A minimal training strategy to play Flappy Bird indefinitely with NEAT.

SBGames, pp. 21–28.

Selected Accreditations & Courses

Databricks: Academy Accreditation – Databricks Fundamentals.

NVIDIA: Fundamentals of Deep Learning.

DataCamp: Data Analysis in Databricks (track); Credit Risk Modeling in Python.